



NORTH SYDNEY BOYS HIGH SCHOOL

2020 YEAR 12 ASSESSMENT TASK 3

Mathematics Advanced

General Instructions

- Working time – 3 hours
- **Reading time – 10 minutes**
- Write on the lined paper in the booklet provided
- Write Multiple Choice responses on sheet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

- Use the Reference Sheet

Class Teacher: Please shade the circle.

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Ms Lee
- ☐ Mr Umakanthan

STUDENT NUMBER: 4.....

(To be used by the exam markers only.)

Questions	1-10	11-17	18-21	22-25	26-29	30-34	35-38	Total
Mark	$\overline{10}$	$\overline{15}$	$\overline{16}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{14}$	$\overline{100}$

Section I

Use the multiple-choice answer sheet for Questions 1 – 10

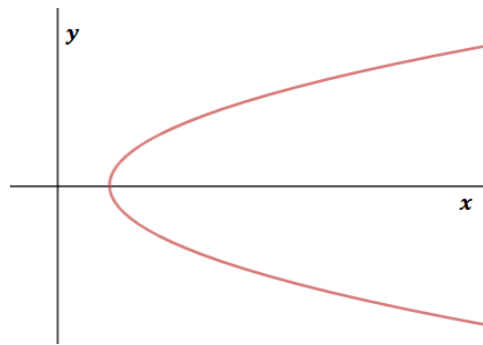
1. The period of $f(x) = 4 \cos\left(\frac{2\pi x}{3}\right) + 1$ is:

- A. $\frac{2}{3}$
- B. $\frac{3}{2}$
- C. 3
- D. 5

2. If $f(x) = \frac{x^2+1}{4-x^2}$ then the vertical and horizontal asymptotes are respectively:

- A. $x = -2, \quad x = 2, \quad y = \frac{1}{4}$
- B. $x = -2, \quad x = 2, \quad y = -1$
- C. $x = -2, \quad x = 2, \quad y = 1$
- D. $x = -2, \quad x = 2, \quad y = 0$

3. The graph below shows the relation $y^2 = x - 1$. What type of relation is it?



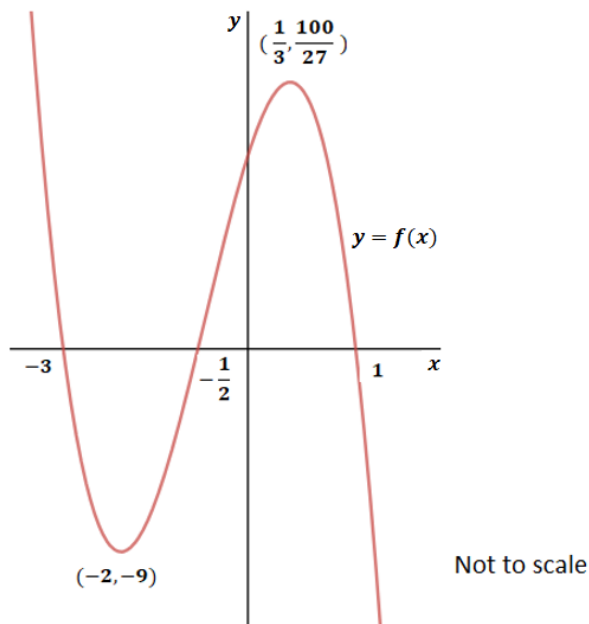
- A. one-to-one
- B. many-to-one
- C. many-to-many
- D. one-to-many

4. The 7th term of an arithmetic sequence is 45 and the 11th term is 77.

Find the first term (a) and the common difference (d).

- A. $a = -3$ and $d = 8$
- B. $a = 3$ and $d = 8$
- C. $a = 8$ and $d = -3$
- D. $a = 8$ and $d = 3$

5. Part of the graph $y = f(x)$ of the function f is shown:



$f'(x) < 0$ for

- A. $x \in (-3, -\frac{1}{2}) \cup (1, \infty)$
- B. $x \in (-9, \frac{100}{27})$
- C. $x \in (-2, \frac{1}{3})$
- D. $x \in (-\infty, -2) \cup (\frac{1}{3}, \infty)$

6. Find the derivative of $e^{x \sin 3x}$.

- A. $e^{3x \cos 3x}$
- B. $e^{x \sin 3x}(\sin 3x + 3x \cos 3x)$
- C. $e^{x \sin 3x}$
- D. $e^{x \sin 3x}(\sin 3x - 3x \cos 3x)$

7. The graph of a function f is obtained from the graph of the function g with rule $g(x) = \sqrt{2x - 5}$ by a reflection in the x -axis, followed by a horizontal dilation from the y -axis by a factor of $\frac{1}{2}$.

Which one of the following is the rule for the function f ?

- A. $f(x) = \sqrt{5 - 4x}$
- B. $f(x) = -\sqrt{x - 5}$
- C. $f(x) = \sqrt{x + 5}$
- D. $f(x) = -\sqrt{4x - 5}$

8. The discrete random variable X has this probability distribution:

x	0	1	2	3
$P(X=x)$	a	$3a$	$5a$	$7a$

The mean of X is:

- A. $\frac{1}{16}$
- B. 1
- C. $\frac{35}{16}$
- D. $\frac{17}{8}$

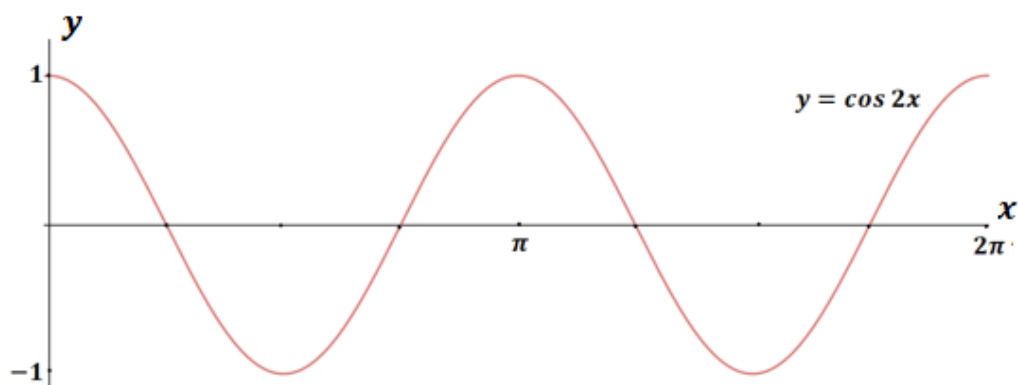
9. The difference in intensity of two sources of sound P_1 and P_2 is defined

to be $10 \log_{10} \left(\frac{P_1}{P_2} \right)$ decibels. How much louder is a sound of 112 dB than a sound of 80 dB ?

- A. 32 times
- B. 1 585 times
- C. 25 times
- D. 3.2 times

10. How many solutions does $6 \cos 2x = x$ for $0 \leq x \leq 2\pi$ have ?

(note: graph below shows $y = \cos 2x$)



- A. 1
- B. 2
- C. 3
- D. 4

Section II: Short Answer

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet. If you use this space, clearly indicate which question you are answering.

Question 11 (1 mark)

Factorise $2x^2 + 5x + 2$

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Question 12 (2 marks)

Rationalise the denominator: $\frac{2}{3-\sqrt{2}}$

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Question 13 (3 marks)

Find the following integrals:

(a) $\int \frac{1-2x^5}{x} dx$

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(b) $\int (3x + 2)^4 dx$

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Question 14 (3 marks)

Differentiate the following functions:

(a) $y = 3^{5x+2}$

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(b) $y = \frac{x}{\log_e x}$

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Question 15 (1 mark)

Write down the domain of $g(x) = \log_e(x + \pi)$

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Question 16 (3 marks)

Solve $2 \log_e x = \log_e(2x + 3)$

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Question 17 (2 marks)

Solve the inequality $x^2 \geq 3x + 18$

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Question 18 (2 marks)

Find $\int_1^{e^3} \frac{5}{x} dx$

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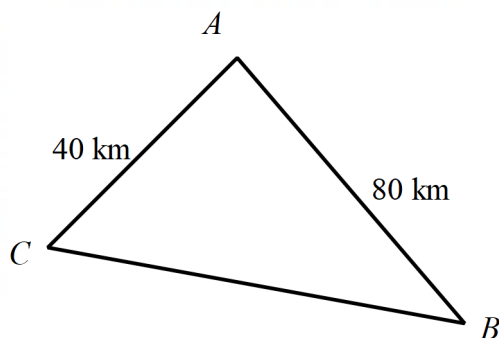
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Question 19 (4 marks)

Three towns, A , B and C form a triangle.

Town A is 80 km from Town B and Town C is 40 km from Town A as shown below:



The bearing of Town B from Town A is 130° . The bearing of Town C from Town A is 240°

- (a) Use this information to find the size of $\angle CAB$, and hence find the area of the triangle formed by the three towns to the nearest square kilometre.

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- (b) Using the cosine rule, find the distance between Town B and Town C , to the nearest kilometre.

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Question 20 (3 marks)

- (a) Given $f(x) = \sqrt{4 - x^2}$ complete this table of values, correct to 3 decimal places. 1

x	0	0.5	1	1.5	2
$f(x)$					

- (b) Use the Trapezoidal rule, with four sub-intervals, to estimate the value of

$$\int_0^2 \sqrt{4 - x^2} dx.$$

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Question 21 (7 marks)

For the curve: $y = x^3 - 3x^2 - 9x + 4$

(a) Find any stationary points and determine their nature.

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(b) Find any points of inflexion.

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(c) Sketch the curve, showing all main features.

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Question 22 (2 marks)

Find the exact value of $\cot \theta$ given that $\cos \theta = 0.6$ and $\sin \theta < 0$.

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Question 23 (3 marks)

A geometric progression has 5th term 9 and 13th term 59 049.

(a) Find the first term and the common ratio.

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(b) Find the 19th term.

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Question 24 (5 marks)

The number of bacteria in a culture can be modelled by $B = 120\,000 e^{0.4 t}$
where t is the time in hours after the experiment started.

(a) How many bacteria are there after 6 hours have passed? **1**

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(b) How fast was the culture growing after 6 hours? **1**

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(c) What was the average rate of increase over the first 6 hours? **1**

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(d) How long, in hours and minutes, will it take until the number of
bacteria doubles? **2**

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Question 25 (5 marks)

In an experiment, 2 balls are drawn at random and without replacement from an urn containing 4 red balls and 6 black balls. Let X be the number of red balls selected.

(a) Complete the table below:

2

Outcome	RR	RB	BR	BB	
X	2	1	1	0	
$p(X = x)$	$\frac{2}{15}$				
$x \cdot p(x)$					
X^2					

(b) What is the expected number of red balls drawn?

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(c) What is the variance, $V(X)$, of this distribution?

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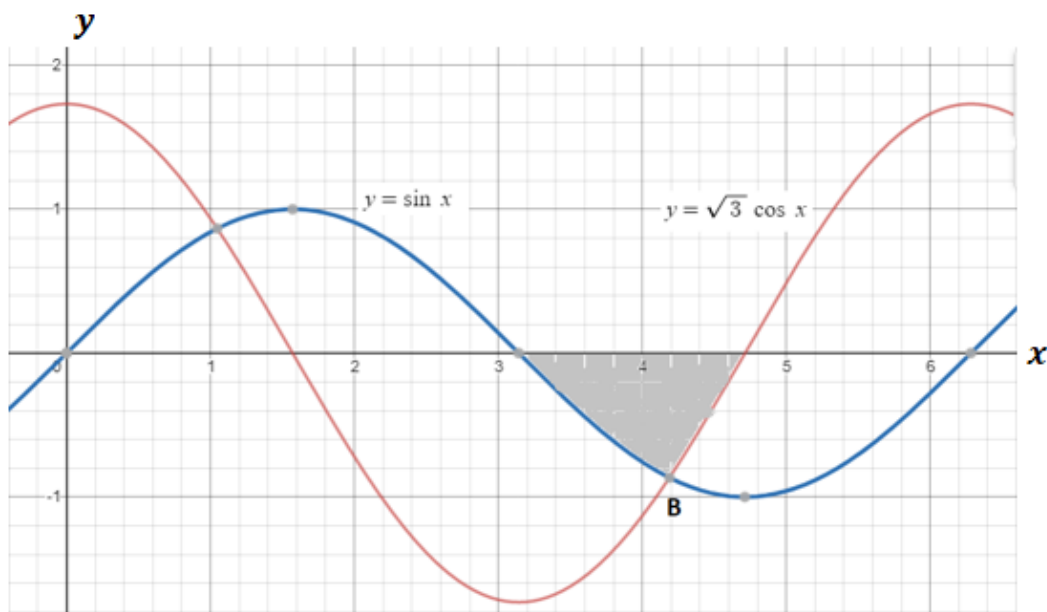
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Question 26 (4 marks)



The diagram above shows the graphs of $y = \sin x$ and $y = \sqrt{3} \cos x$, $0 \leq x \leq 2\pi$.
The second point of intersection is labelled B .

- (a) Show, using any appropriate method, that B has coordinates $\left(\frac{4\pi}{3}, \frac{-\sqrt{3}}{2}\right)$

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- (b) Find the exact area of the shaded region.

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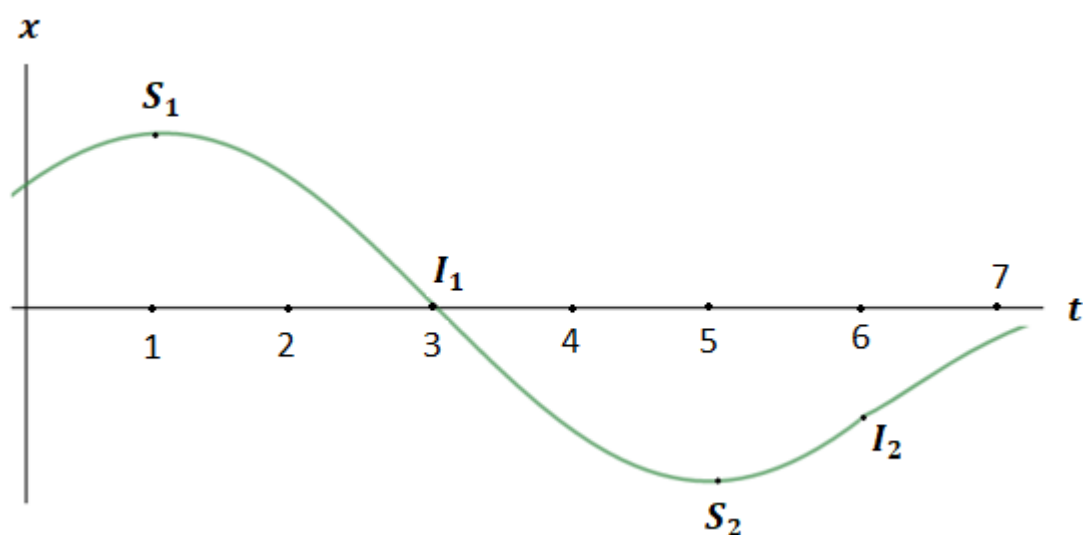
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Question 27 (3 marks)



The graph shows the displacement of a particle, moving in a straight line, over the first 7 seconds of its motion. S_1 and S_2 are stationary points, and I_1 and I_2 are inflection points.

State the times, or periods of time, for which:

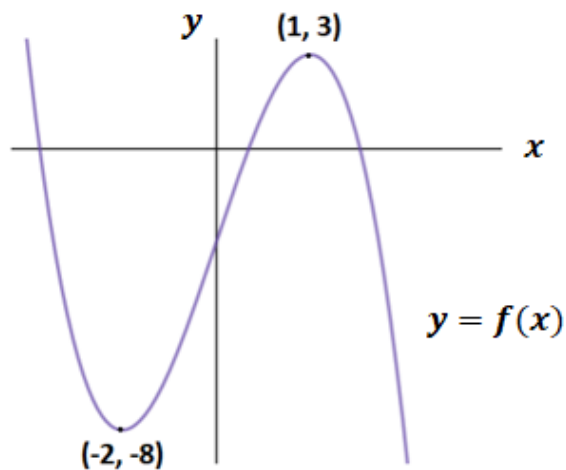
(a) The particle is stationary. 1

(b) The velocity is negative. 1

(c) The acceleration is positive. 1

Question 28 (4 marks)

Consider the graph of $y = f(x)$ shown:



- (a) Use the space below to sketch the graph of $y = f'(x)$

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- (b) Find the area bounded by $y = f'(x)$ and the x-axis.

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Question 29 (4 marks)

For events A and B from a sample space, $P(A|B) = \frac{3}{4}$ and $P(B) = \frac{1}{3}$

(a) Calculate $P(A \cap B)$

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(b) Calculate $P(\bar{A} \cap B)$ where $\bar{A}$ denotes the complement of A.

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(c) If A and B are independent, calculate $P(A \cup B)$

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Question 30 (2 marks)

The gradient of a curve is given by $\frac{dy}{dx} = \frac{3x}{x^2+e}$

The curve passes through (0, 2). What is its equation?

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Question 31 (3 marks)

If $f(x) = \sqrt{2-x}$ and $g(x) = \sqrt{x}$, then

(a) Find the rule for the composite function $f \circ g$

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(b) Find the domain of $f \circ g$

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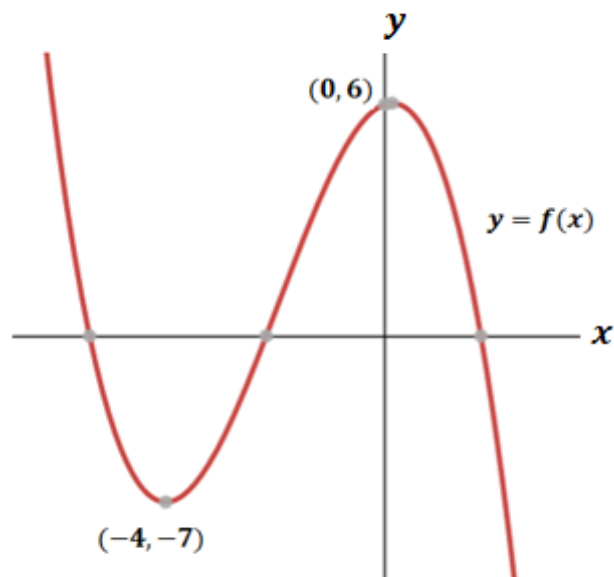
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Question 32 (2 marks)

Given the graph of the function $y = f(x)$ below, with turning points as shown, sketch the transformed function $y = 3f(x + 2) - 4$. (x-intercepts not required).

2



Question 33 (5 marks)

- (a) Differentiate $y = \log_e (\cos x)$ with respect to x .

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- (b) Sketch $y = \tan x$ for $0 \leq x \leq \frac{\pi}{2}$

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- (c) Hence, using parts (a) and (b), find the area bounded by $y = \tan x$, the x -axis, and the line $x = \frac{\pi}{3}$
(leave answer in simplest exact form)

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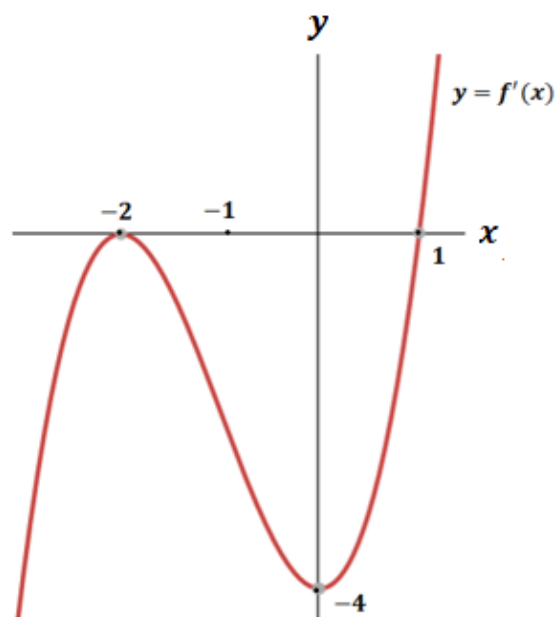
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Question 34 (3 marks)

The diagram shows $y = f'(x)$, the graph of the derivative function of $y = f(x)$.



- (a) Explain why there is a horizontal point of inflection at $x = -2$

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- (b) Given that $f(0) = 2$, sketch a possible graph of $y = f(x)$.

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Question 35 (3 marks)

Find the equation of the normal to $y = x \sin x$ at the point where $x = \frac{\pi}{2}$.

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Question 36 (2 marks)

(a) Sketch $y = |x| - 1$ and $y = 2x + 2$ neatly on the same number plane.

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(b) Hence solve the equation $|x| - 2x = 3$

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Question 37 (4 marks)

One half percent (0.5 %) of a country has a certain viral disease. A test is developed for the disease. The test gives a false positive 3% of the time, and a false negative 2% of the time.

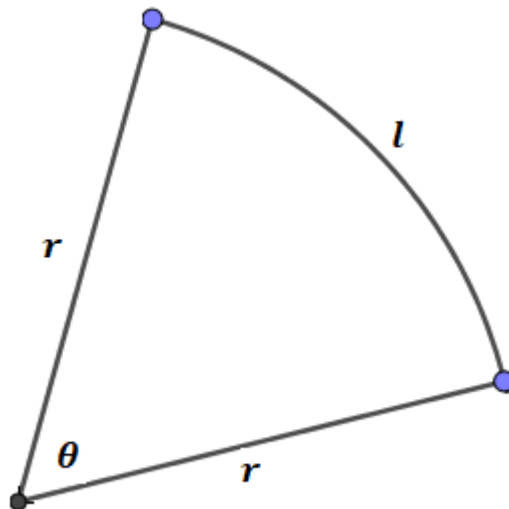
- (a) Show that the probability that Andy, a randomly selected person, tests positive is 0.03475 2

[Hint: in this question, let D be the event that Andy has the disease, and $\bar{D}$ be the event Andy does not have it. Let T be the event that Andy's test comes back positive.]

- (b) Andy just got the bad news that his test came back positive.
Find the probability that Andy actually has the disease.2

Question 38 (5 marks)

The diagram below shows a sector of a circle of radius r centimetres. The angle at the centre is θ radians, and the perimeter of the whole sector is 8 cm .



- (a) Show that $r = \frac{8}{2+\theta}$.

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- (b) Show that A , the area of the sector in cm^2 , is given by

$$A = \frac{32\theta}{(\theta+2)^2}$$

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[illegible]

26

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g "●"

STUDENT NUMBER:

Class (please ✓)

☐ - Mr Ireland

☐ - Dr Jomaa

☐ - Miss Lee

☐ - Mr Lin

☐ - Mrs Sarofim

☐ - Ms Ziazaris

☐ - Mr Berry

☐ - Mr Hwang

☐ - Mr Uma

1 -	☐ A	☐ B	☒ C	☐ D
2 -	☐ A	☒ B	☐ C	☐ D
3 -	☐ A	☐ B	☐ C	☒ D
4 -	☒ A	☐ B	☐ C	☐ D
5 -	☐ A	☐ B	☐ C	☒ D
6 -	☐ A	☒ B	☐ C	☐ D
7 -	☐ A	☐ B	☐ C	☒ D
8 -	☐ A	☐ B	☐ C	☒ D
9 -	☐ A	☒ B	☐ C	☐ D
10 -	☐ A	☐ B	☒ C	☐ D

2020 Mathematics Advanced - Task 3

$$(1) \quad T = \frac{2\pi}{\frac{2\pi}{3}} = 3 \quad \therefore (C)$$

$$(2) \quad f(x) = \frac{x^2 + 1}{4 - x^2} \quad \therefore x \neq \pm 2 \quad \therefore (B)$$

as $x \rightarrow \infty$, $y \rightarrow -1$

$$(3) \quad \text{one-to-many,} \quad \therefore (D)$$

$$(4) \quad T_7 = a + 6d = 45$$

$$T_{11} = a + 10d = 77$$

$$\therefore 4d = 32, \quad d = 8$$

$$\therefore a = -3 \quad \therefore (A)$$

$$(5) \quad f'(x) < 0 \text{ means gradient}$$

is negative, $\therefore x < -2$ $\therefore (D)$

or $x > \frac{1}{3}$

$$(6) \quad \frac{d}{dx} e^{x \sin 3x} = (\sin 3x \cdot 1 + x \cdot 3 \cos 3x) \cdot e^{x \sin 3x}$$

$$\therefore (B)$$

$$(7) \quad \sqrt{2x-5} \rightarrow -\sqrt{2x-5} \rightarrow -\sqrt{2\left(\frac{x}{2}\right)-5}$$

$$= -\sqrt{4x-5} \quad \therefore (D)$$

$$\textcircled{8} \quad \sum p = 1 \therefore a + 3a + 5a + 7a = 1$$

$$\therefore a = \frac{1}{16}$$

$$\mu = \sum x \cdot p(x) = 0 + 3a + 10a + 21a$$

$$= 34a$$

$$= \frac{34}{16} = \frac{17}{8} \therefore \textcircled{D}$$

$$\textcircled{9} \quad 112 \text{ dB} - 80 \text{ dB} = 32 \text{ dB}$$

$$\therefore \log_{10} \left(\frac{P_1}{P_2} \right) = 3.2 \therefore \frac{P_1}{P_2} = 10^{3.2}$$

$$\doteq 1585$$

$\therefore \textcircled{B}$

$$\textcircled{10} \quad 6 \cos 2x = x$$

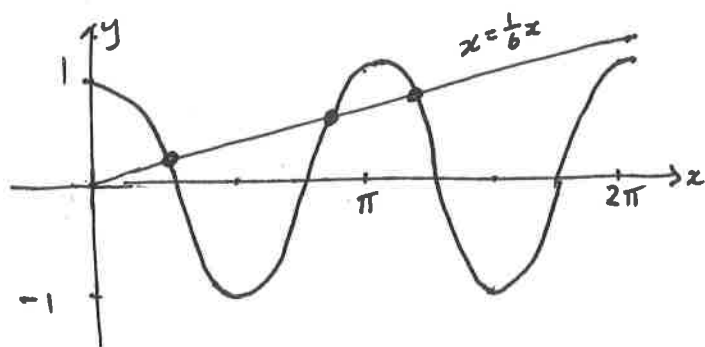
means $\cos 2x = \frac{1}{6}x$.

If we draw $y = \frac{1}{6}x$, then at

$$x = 2\pi, y \doteq 1.047 > \cos 2(2\pi)$$

$\therefore$ 3 intersections from $x=0$ to $x=2\pi$

$\therefore \textcircled{C}$



Section II: Short Answer

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet.
If you use this space, clearly indicate which question you are answering.

Question 11 (1 mark)

Factorise $2x^2 + 5x + 2$

1

$$(2x+1)(x+2)$$

Question 12 (2 marks)

Rationalise the denominator: $\frac{2}{3-\sqrt{2}}$

2

$$= \frac{2}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}}$$

$$= \frac{6 + 2\sqrt{2}}{9-2}$$

$$= \frac{6 + 2\sqrt{2}}{7}$$

Question 13 (3 marks)

Find the following integrals:

(a) $\int \frac{1-2x^5}{x} dx$

2

$$= \int \left(\frac{1}{x} - 2x^4 \right) dx$$

$$= \ln x - \frac{2x^5}{5} + C$$

(b) $\int (3x+2)^4 dx$

1

=

$$\frac{(3x+2)^5}{5} + C$$

✓

Question 14 (3 marks)

Differentiate the following functions:

(a) $y = 3^{5x+2}$

1

$$y' = \ln 3 \cdot 5 \cdot 3^{5x+2}$$

i.e. $y' = 5 \ln 3 \cdot 3^{5x+2}$

✓

(b) $y = \frac{x}{\log_e x}$

2

$$y' = \frac{\log x \cdot 1 - x \cdot \frac{1}{x}}{(\log x)^2}$$

✓

$$\therefore y' = \frac{\log_e x - 1}{(\log_e x)^2}$$

✓

Question 15 (1 mark)

Write down the domain of $g(x) = \log_e(x + \pi)$

1

We need $x + \pi > 0 \therefore x > -\pi$

$$\therefore \mathcal{D} : x > -\pi$$

✓

Question 16 (3 marks)Solve $2 \log_e x = \log_e(2x + 3)$

3

$$\log_e x^2 = \log_e(2x+3)$$

$$\therefore x^2 = 2x + 3$$

$$\therefore x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$\therefore x = 3 \text{ or } x = -1$$

But $x \neq -1$ as this can't be subbed into the equation

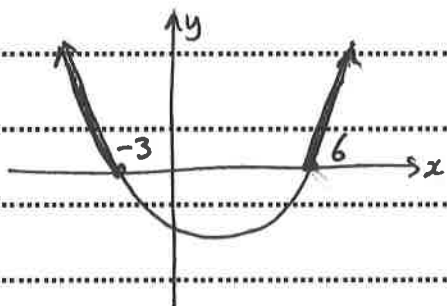
$$\therefore \boxed{x = 3}$$

Question 17 (2 marks)Solve the inequality $x^2 \geq 3x + 18$

2

$$x^2 - 3x - 18 \geq 0$$

$$(x-6)(x+3) \geq 0$$



$$\therefore \boxed{x \leq -3 \text{ or } x \geq 6}$$

Question 18 (2 marks)Find $\int_1^{e^3} \frac{5}{x} dx$

2

$$= [5 \ln x]_1^{e^3}$$

$$= 5 \ln e^3 - 5 \ln 1$$

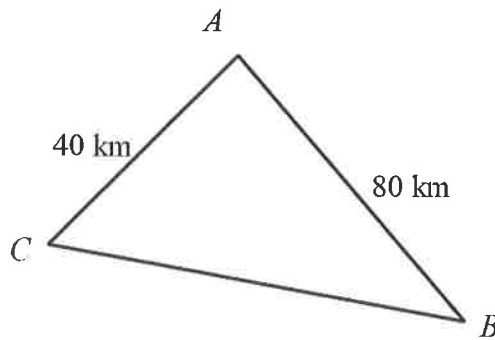
$$= 5 \times 3 - 5 \times 0$$

$$= \boxed{15}$$

Question 19 (4 marks)

Three towns, A , B and C form a triangle.

Town A is 80 km from Town B and Town C is 40 km from Town A as shown below:



The bearing of Town B from Town A is 130° . The bearing of Town C from Town A is 240°

- (a) Use this information to find the size of $\angle CAB$, and hence find the area of the triangle formed by the three towns to the nearest square kilometre.

2

$$\angle CAB = 240^\circ - 130^\circ = 110^\circ$$

$$\therefore \text{Area } \triangle ABC = \frac{1}{2} \cdot 40 \cdot 80 \cdot \sin 110^\circ$$

$$\doteq 1503.508193 \dots$$

$$= 1504 \text{ km}^2 \text{ (nearest km}^2\text{)}$$

✓✓

- (b) Using the cosine rule, find the distance between Town B and Town C , to the nearest kilometre.

2

$$BC^2 = 40^2 + 80^2 - 2(40)(80) \cos 110^\circ$$

✓

$$\doteq 10188.92892 \dots$$

$$\therefore BC \doteq 100.940224 \dots$$

$$\therefore BC = 101 \text{ km (nearest km)}$$

✓

Question 20 (3 marks)

- (a) Given $f(x) = \sqrt{4-x^2}$ complete this table of values, correct to 3 decimal places. 1

x	0	0.5	1	1.5	2
$f(x)$	2	1.936	1.732	1.323	0

- (b) Use the Trapezoidal rule, with four sub-intervals, to estimate the value of

$$\int_0^2 \sqrt{4-x^2} dx.$$

2

$h = \text{width sub-interval} = 0.5$

$$\int_0^2 \sqrt{4-x^2} dx \div \frac{0.5}{2} [2 + 0 + 2(1.936 + 1.732 + 1.323)]$$

$$= 2.9955$$

Question 21 (7 marks)

For the curve: $y = x^3 - 3x^2 - 9x + 4$

(a) Find any stationary points and determine their nature.

3

$$y' = 3x^2 - 6x - 9 \quad ; \quad y'' = 6x - 6$$
$$= 3(x^2 - 2x - 3) \quad \therefore y'' = 6(x-1)$$

✓

$$\therefore y' = 3(x-3)(x+1)$$

For stat. pts, $y' = 0 \therefore x = 3 \quad \text{or} \quad x = -1$

$$y = -23 \quad \quad \quad y = 9$$

✓

At $(3, -23)$, $y'' = 6(3-1) = 12 > 0 \therefore$ local minimum
at $(3, -23)$

At $(-1, 9)$, $y'' = 6(-1-1) = -12 < 0 \therefore$ local maximum
at $(-1, 9)$

✓

(b) Find any points of inflexion.

2

For inflexions, $y'' = 0$

$$\therefore 6(x-1) = 0 \therefore x = 1, y = -7.$$

✓

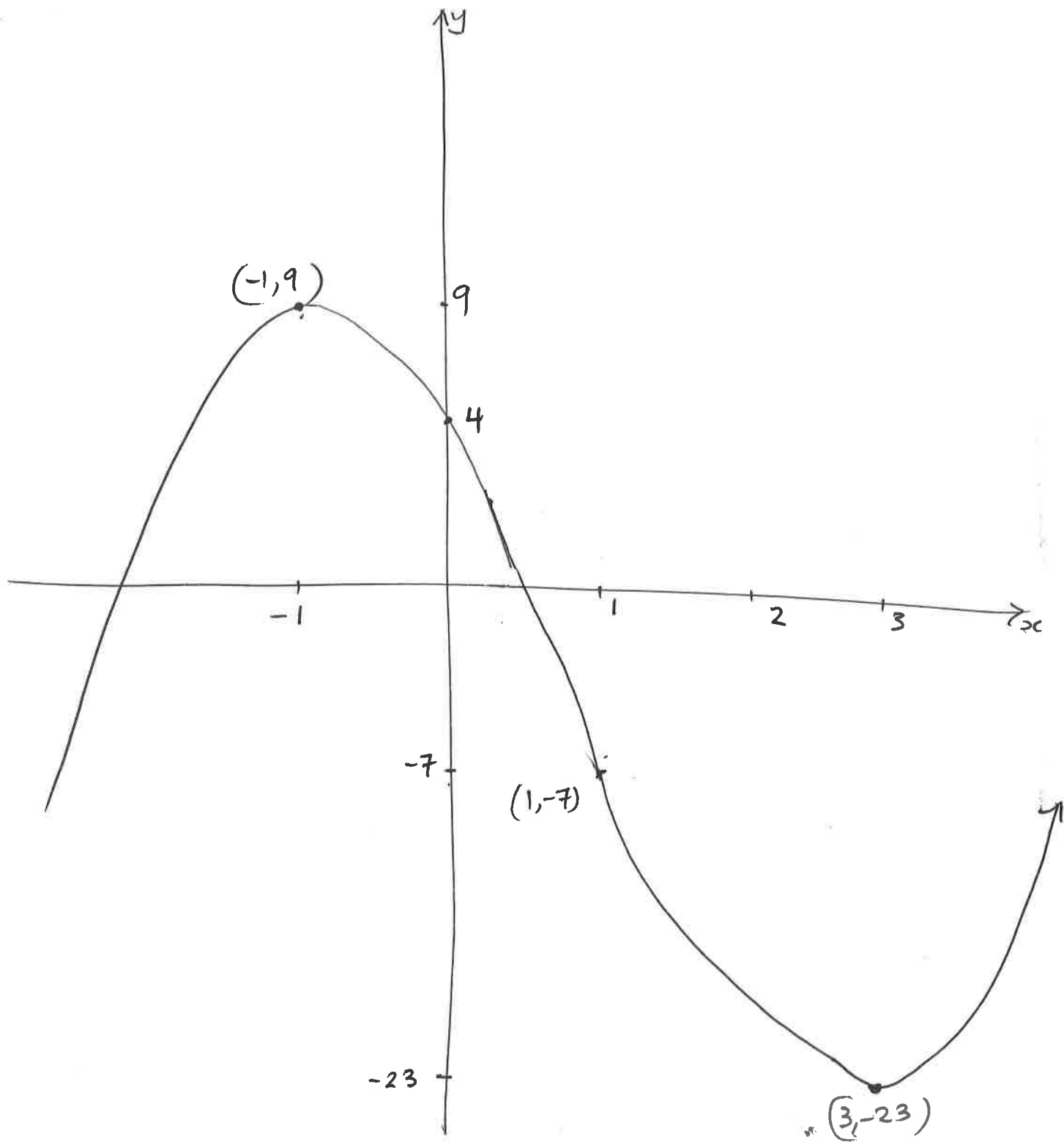
Test :

x	0	1	2
y''	-6	0	6

change in concavity, $\therefore$ inflexion at $(1, -7)$ ✓

(c) Sketch the curve, showing all main features.

2



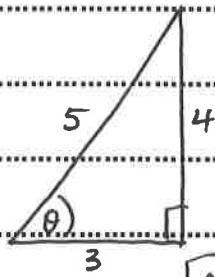
✓✓

Question 22 (2 marks)Find the exact value of $\cot \theta$ given that $\cos \theta = 0.6$ and $\sin \theta < 0$.

2

$$\cos \theta = 0.6 = \frac{3}{5}; \quad \cos \theta > 0 \text{ \& \; } \sin \theta < 0 \therefore Q4$$

$$\therefore \tan \theta < 0.$$



$$\therefore \tan \theta = -\frac{4}{3}$$

$$\therefore \cot \theta = -\frac{3}{4}$$

✓✓

$$[\text{ALT: } \sin \theta = -\sqrt{1 - \cos^2 \theta}, \text{ as in Q4} = -\sqrt{1 - \frac{9}{25}}$$

$$\therefore \sin \theta = -\frac{4}{5} \therefore \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\frac{3}{5}}{-\frac{4}{5}} = -\frac{3}{4}]$$

Question 23 (3 marks)A geometric progression has 5th term 9 and 13th term 59 049.

(a) Find the first term and the common ratio.

2

$$T_5 = ar^4 = 9$$

$$T_{13} = ar^{12} = 59049$$

$$\therefore \frac{ar^{12}}{ar^4} = \frac{59049}{9}$$

$$\therefore r^8 = 6561$$

$$\therefore r = 3 \text{ or } -3$$

$$\therefore a(81) = 9 \therefore a = \frac{1}{9}$$

✓✓

(b) Find the 19th term.

1

$$T_{19} = ar^{18} = \frac{1}{9} \cdot (3)^{18} \quad \left(\text{or } \frac{1}{9} (-3)^{18} \right)$$

$$= 3^{16}$$

$$= 43\,046\,721$$

✓

Question 24 (5 marks)

The number of bacteria in a culture can be modelled by $B = 120\,000 e^{0.4t}$

where t is the time in hours after the experiment started.

- (a) How many bacteria are there after 6 hours have passed?

1

$$B = 120\,000 e^{(0.4)6} = 120\,000 e^{2.4} \\ = 1\,322\,781$$

- (b) How fast was the culture growing after 6 hours?

1

$$\frac{dB}{dt} = 0.4 \times 120\,000 e^{0.4t} = 48\,000 e^{0.4t}$$

$$\therefore \text{at } t=6, \frac{dB}{dt} = 48\,000 e^{2.4} \\ = 529\,112 \text{ per hour.}$$

- (c) What was the average rate of increase over the first 6 hours?

1

$$\text{Starting number at } t=0 = 120\,000 e^0 = 120\,000$$

$$\therefore \text{average rate} = \frac{1\,322\,781 - 120\,000}{6}$$

$$= 200\,463 \text{ per hour}$$

- (d) How long, in hours and minutes, will it take until the number of bacteria doubles?

2

$$240\,000 = 120\,000 e^{0.4t}$$

$$\therefore e^{0.4t} = 2$$

$$\therefore 0.4t = \ln 2$$

$$\therefore t = \frac{\ln 2}{0.4} \doteq 1.732\,867... \text{ hours}$$

$$= 1 \text{ hour } 44 \text{ mins. (nearest min.)}$$

Question 25 (5 marks)

In an experiment, 2 balls are drawn at random and without replacement from an urn containing 4 red balls and 6 black balls. Let X be the number of red balls selected.

(a) Complete the table below:

2

Outcome	RR	RB	BR	BB	
X	2	1	1	0	
$p(X = x)$	$\frac{2}{15}$	$\frac{4}{10} \cdot \frac{6}{9} = \frac{4}{15}$	$\frac{6}{10} \cdot \frac{4}{9} = \frac{4}{15}$	$\frac{6}{10} \cdot \frac{5}{9} = \frac{1}{3}$	
$x \cdot p(x)$	$\frac{4}{15}$	$\frac{4}{15}$	$\frac{4}{15}$	4 0	
X^2	4	1	1	0	

✓✓

(b) What is the expected number of red balls drawn?

1

$$E(X) = \sum x \cdot p(x)$$

$$= \frac{4}{15} + \frac{4}{15} + \frac{4}{15} + 0 = \frac{12}{15}$$

$$= 0.8$$

✓

(c) What is the variance, $V(X)$, of this distribution?

2

$$E(X^2) = 4 \cdot \frac{2}{15} + 1 \cdot \frac{4}{15} + 1 \cdot \frac{4}{15} + 0 \cdot \frac{1}{3} = \frac{16}{15}$$

$$\therefore V(X) = E(X^2) - (E(X))^2$$

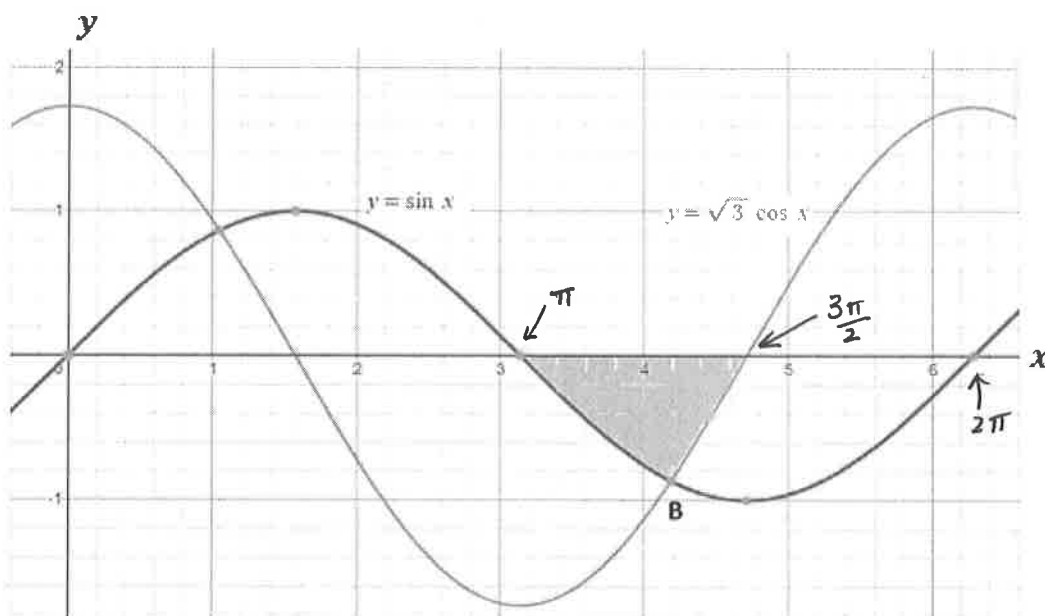
$$= \frac{16}{15} - (0.8)^2$$

$$= \frac{32}{75}$$

$$(= 0.427 \text{ 3d.p.})$$

✓✓

Question 26 (4 marks)



The diagram above shows the graphs of $y = \sin x$ and $y = \sqrt{3} \cos x$, $0 \leq x \leq 2\pi$. The second point of intersection is labelled B.

- (a) Show, using any appropriate method, that B has coordinates $(\frac{4\pi}{3}, -\frac{\sqrt{3}}{2})$ 1

B lies on $\sin x$ and on $\sqrt{3} \cos x$.
 Here, $\sin \frac{4\pi}{3} = -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}$
 and $\sqrt{3} \cos \frac{4\pi}{3} = \sqrt{3} \cdot -\cos \frac{\pi}{3}$
 $= \sqrt{3} \cdot -\frac{1}{2} = -\frac{\sqrt{3}}{2}$
 $\therefore$ curves both go thru $(\frac{4\pi}{3}, -\frac{\sqrt{3}}{2})$

ALT: $\sin x = \sqrt{3} \cos x$
 $\therefore \tan x = \sqrt{3}$
 $\therefore x = \frac{\pi}{3}, \frac{4\pi}{3}$
 But $x > \pi \therefore x = \frac{4\pi}{3}$
 $\therefore y = \sin \frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$

- (b) Find the exact area of the shaded region. 3

Area = $\left| \int_{\pi}^{\frac{4\pi}{3}} \sin x \, dx \right| + \left| \int_{\frac{4\pi}{3}}^{\frac{3\pi}{2}} \sqrt{3} \cos x \, dx \right|$

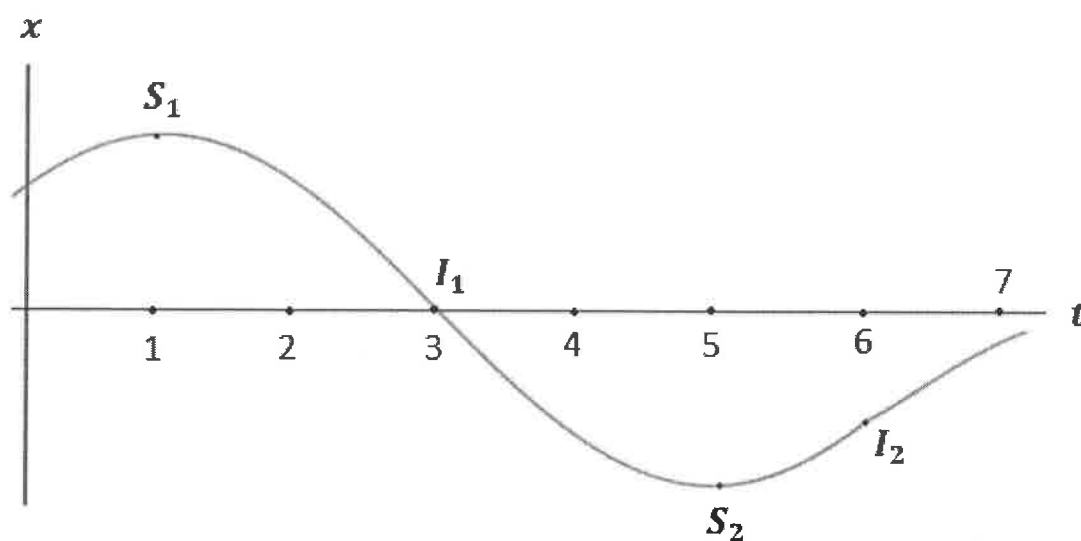
$= \left| \left[-\cos x \right]_{\pi}^{\frac{4\pi}{3}} \right| + \left| \left[\sqrt{3} \sin x \right]_{\frac{4\pi}{3}}^{\frac{3\pi}{2}} \right|$

$= \left| -\cos \frac{4\pi}{3} - (-\cos \pi) \right| + \left| \sqrt{3} \sin \frac{3\pi}{2} - \sqrt{3} \sin \frac{4\pi}{3} \right|$

$= \left| \frac{1}{2} + (-1) \right| + \left| -\sqrt{3} - \sqrt{3} \left(-\frac{\sqrt{3}}{2} \right) \right|$

$= \frac{1}{2} + \left| -\sqrt{3} + \frac{3}{2} \right| = \frac{1}{2} + \sqrt{3} - \frac{3}{2} = (\sqrt{3} - 1) \text{ units}^2$

Question 27 (3 marks)



The graph shows the displacement of a particle, moving in a straight line, over the first 7 seconds of its motion. S_1 and S_2 are stationary points, and I_1 and I_2 are inflection points.

State the times, or periods of time, for which:

(a) The particle is stationary.

At $t=1$ and $t=5$

1

(b) The velocity is negative.

$$v = \frac{dx}{dt} \therefore$$

$1 < t < 5$

1

(c) The acceleration is positive.

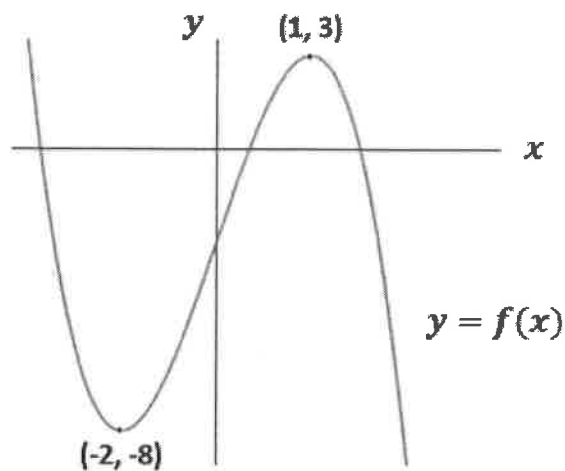
$$a = \frac{d^2x}{dt^2} \therefore$$

concave up $\therefore 3 < t < 6$

1

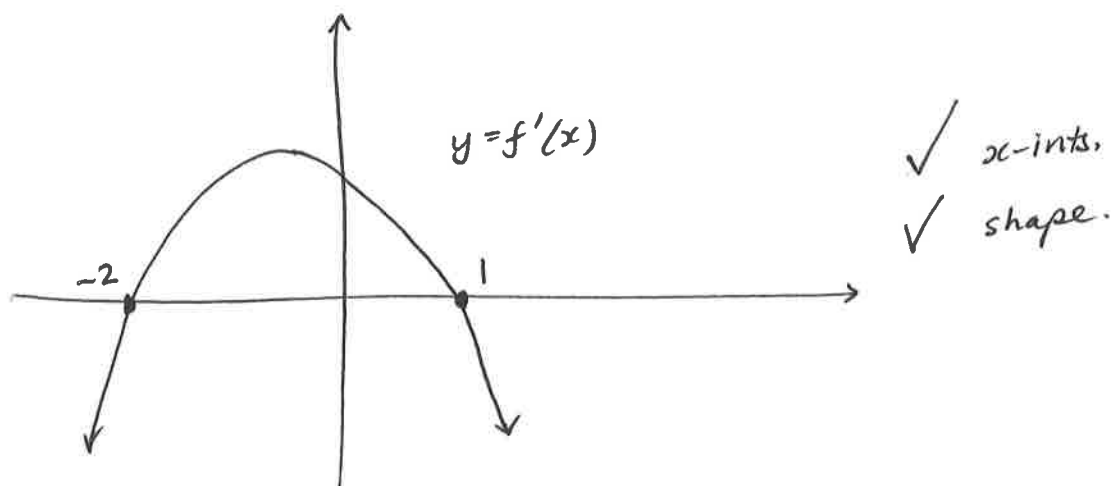
Question 28 (4 marks)

Consider the graph of $y = f(x)$ shown:



- (a) Use the space below to sketch the graph of $y = f'(x)$

2



- (b) Find the area bounded by $y = f'(x)$ and the x-axis.

2

$$\text{Area} = \int_{-2}^1 f'(x) \, dx$$

$$= [f(x)]_{-2}^1$$

$$= 3 - (-8) = 11$$

✓✓

Question 29 (4 marks)

For events A and B from a sample space, $P(A|B) = \frac{3}{4}$ and $P(B) = \frac{1}{3}$

(a) Calculate $P(A \cap B)$

1

$$\begin{aligned} P(A \cap B) &= P(A|B) \cdot P(B) \\ &= \frac{3}{4} \cdot \frac{1}{3} \\ &= \left(\frac{1}{4} \right) \end{aligned}$$

(b) Calculate $P(\bar{A} \cap B)$ where $\bar{A}$ denotes the complement of A.

1

$$\begin{aligned} P(\bar{A} \cap B) &= P(\bar{A}|B) \cdot P(B) \\ &= \frac{1}{4} \cdot \frac{1}{3} \\ &= \left(\frac{1}{12} \right) \end{aligned}$$

$$\begin{aligned} \text{[ALT: } P(\bar{A} \cap B) &= P(B) - P(A \cap B) \\ &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

(c) If A and B are independent, calculate $P(A \cup B)$

2

"Independent" means $P(A) = P(A|B) = \frac{3}{4}$

$$\begin{aligned} \therefore P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{3}{4} + \frac{1}{3} - \frac{1}{4} \\ &= \left(\frac{5}{6} \right) \end{aligned}$$

Question 30 (2 marks)

The gradient of a curve is given by $\frac{dy}{dx} = \frac{3x}{x^2+e}$

The curve passes through (0, 2). What is its equation?

2

$$\frac{dy}{dx} = \frac{3x}{x^2+e} = \frac{3}{2} \cdot \frac{2x}{x^2+e}$$

$$\therefore y = \frac{3}{2} \ln(x^2+e) + C$$

At $x=0, y=2 \quad \therefore 2 = \frac{3}{2} \ln(0^2+e) + C$

$$2 = \frac{3}{2} \ln e + C$$

$$= \frac{3}{2} \cdot 1 + C \quad \therefore C = \frac{1}{2}$$

$$\therefore y = \frac{3}{2} \log_e(x^2+e) + \frac{1}{2}$$

Question 31 (3 marks)

If $f(x) = \sqrt{2-x}$ and $g(x) = \sqrt{x}$, then

(a) Find the rule for the composite function $f \circ g$

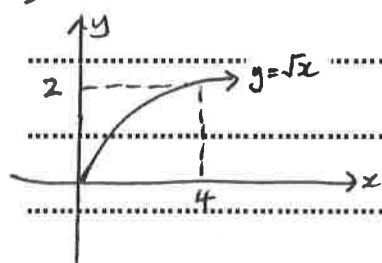
1

$$f \circ g(x) = \sqrt{2-\sqrt{x}}$$

(b) Find the domain of $f \circ g$

2

We need $x \geq 0$ (for $\sqrt{x}$) and also $2-\sqrt{x} \geq 0$.
ie $x \geq 0$ and $\sqrt{x} \leq 2 \quad \therefore x \leq 4$.



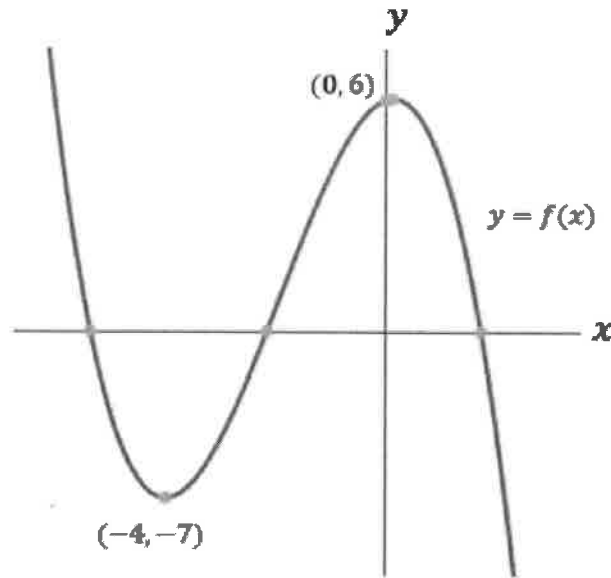
$$\therefore D: 0 \leq x \leq 4$$

$$(ie. [0, 4])$$

Question 32 (2 marks)

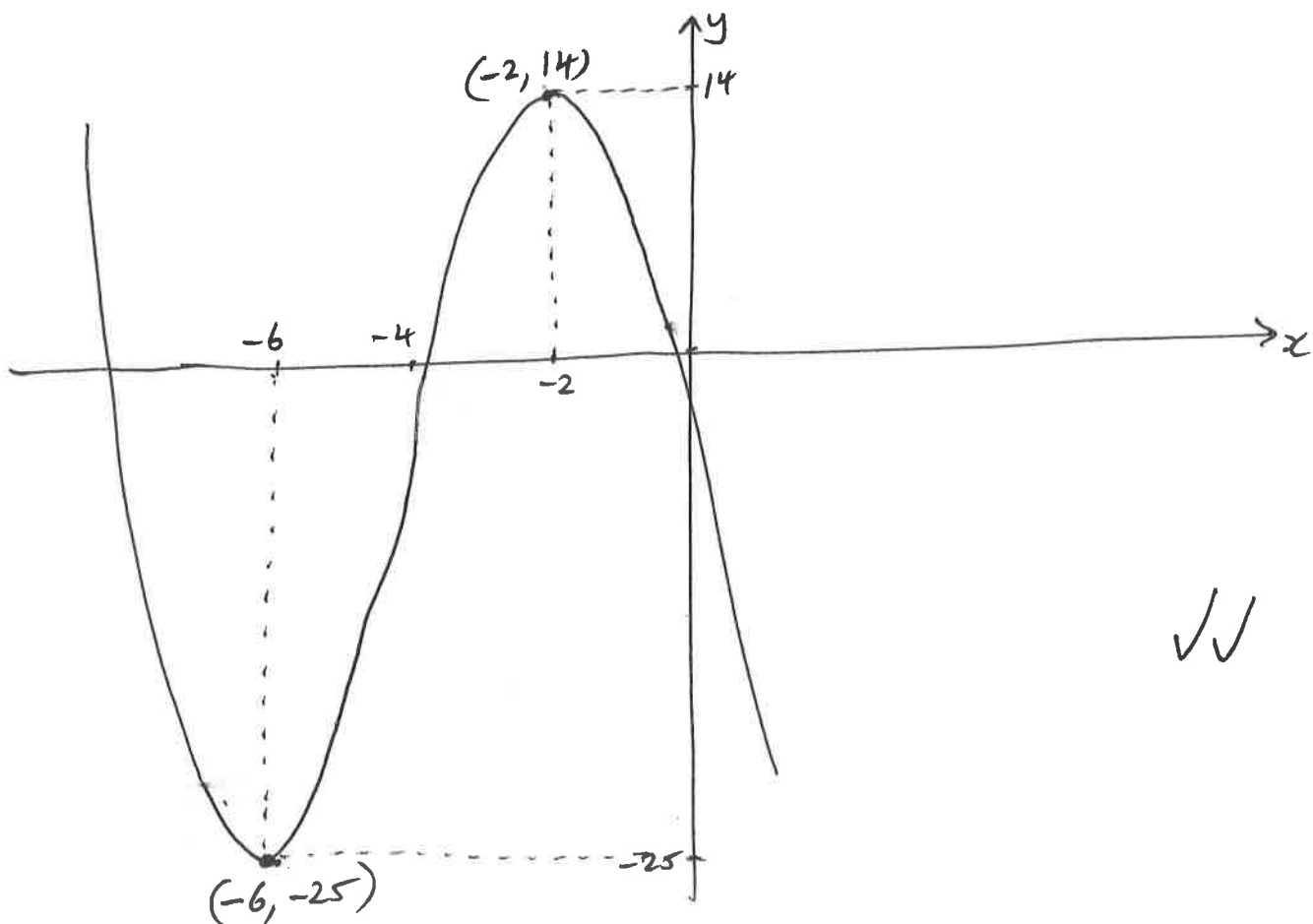
Given the graph of the function $y = f(x)$ below, with turning points as shown, sketch the transformed function $y = 3f(x+2) - 4$. (x-intercepts not required).

2



$(0, 6)$ transforms to $x = -2$
 $y = 3(6) - 4 = 14$ $\therefore (-2, 14)$

$(-4, -7)$ transforms to $x = -6$
 $y = 3(-7) - 4 = -25$ $\therefore (-6, -25)$.



Question 33 (5 marks)

- (a) Differentiate $y = \log_e (\cos x)$ with respect to x .

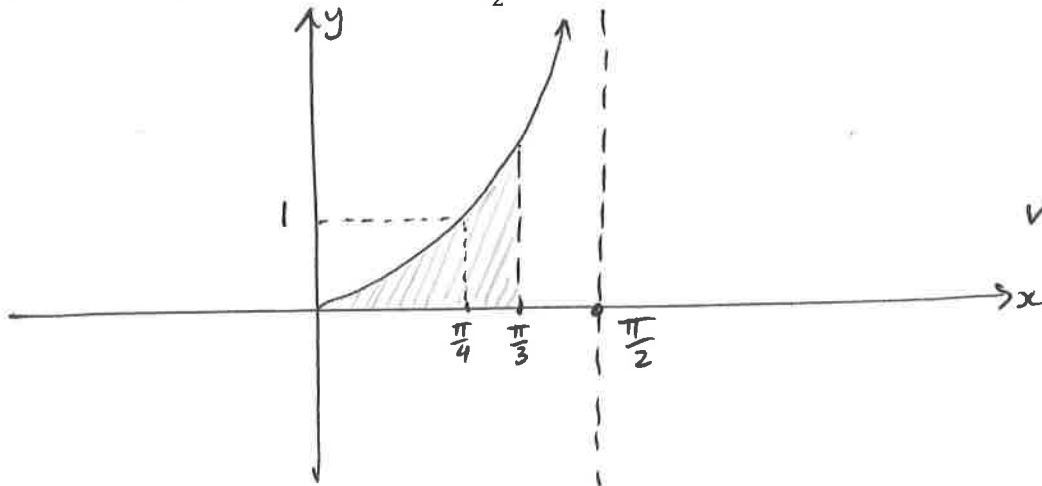
1

$$y' = -\sin x \cdot \frac{1}{\cos x}$$

$$= -\tan x$$

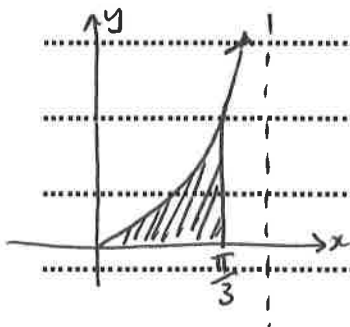
- (b) Sketch $y = \tan x$ for $0 \leq x \leq \frac{\pi}{2}$

1



- (c) Hence, using parts (a) and (b), find the area bounded by $y = \tan x$, the x -axis, and the line $x = \frac{\pi}{3}$ (leave answer in simplest exact form)

3



$$A = \int_0^{\pi/3} \tan x \, dx$$

$$= - \int_0^{\pi/3} -\tan x \, dx$$

$$= - \left[\log_e (\cos x) \right]_0^{\pi/3}$$

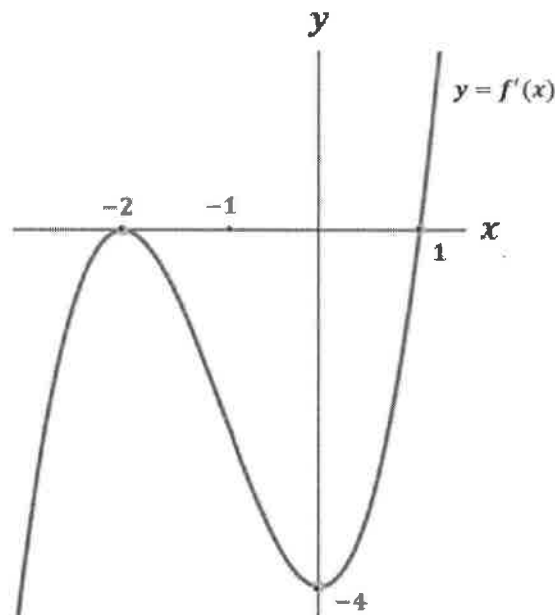
$$= - \left[\log_e \left(\frac{1}{2} \right) - \log_e 1 \right]$$

$$= - \left(\log_e 1 - \log_e 2 - \log_e 1 \right)$$

$$= \log_e 2 \text{ units}^2$$

Question 34 (3 marks)

The diagram shows $y = f'(x)$, the graph of the derivative function of $y = f(x)$.



- (a) Explain why there is a horizontal point of inflection at $x = -2$

1

At $x = -2$, $f'(x) = 0 \therefore$ it's a stationary point.

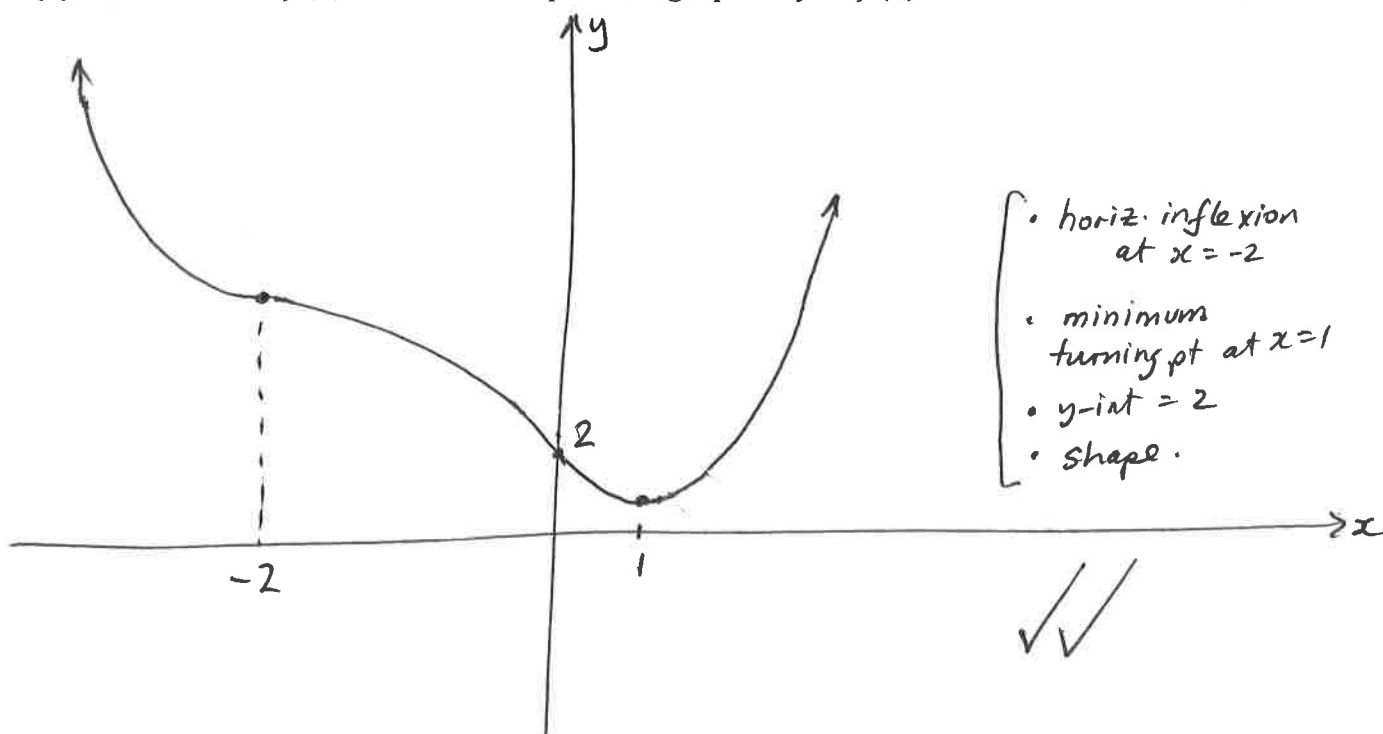
Also, $f''(x) > 0$ on left of $x = -2$, & $f''(x) < 0$ on right

$\therefore$ change in concavity $\therefore$ (horizontal) point of inflection.

(ALT: $f'(x)$ is 0 at $x = -2$ but negative on both sides of -2)
i.e. $\searrow$ $\therefore$ inflection.

- (b) Given that $f(0) = 2$, sketch a possible graph of $y = f(x)$.

2



Question 35 (3 marks)

Find the equation of the normal to $y = x \sin x$ at the point where $x = \frac{\pi}{2}$.

3

$$y = x \sin x \quad \therefore y' = \sin x \cdot 1 + x \cdot \cos x$$

$$\text{i.e. } y' = \sin x + x \cos x$$

$$\text{At } x = \frac{\pi}{2}, \quad y = \frac{\pi}{2} \cdot \sin \frac{\pi}{2} = \frac{\pi}{2}$$

$$\text{and, } y' = \sin \frac{\pi}{2} + \frac{\pi}{2} \cdot \cos \frac{\pi}{2} = 1 \quad \therefore m_N = -1$$

$$\therefore \text{normal is } y - \frac{\pi}{2} = -1 \left(x - \frac{\pi}{2} \right)$$

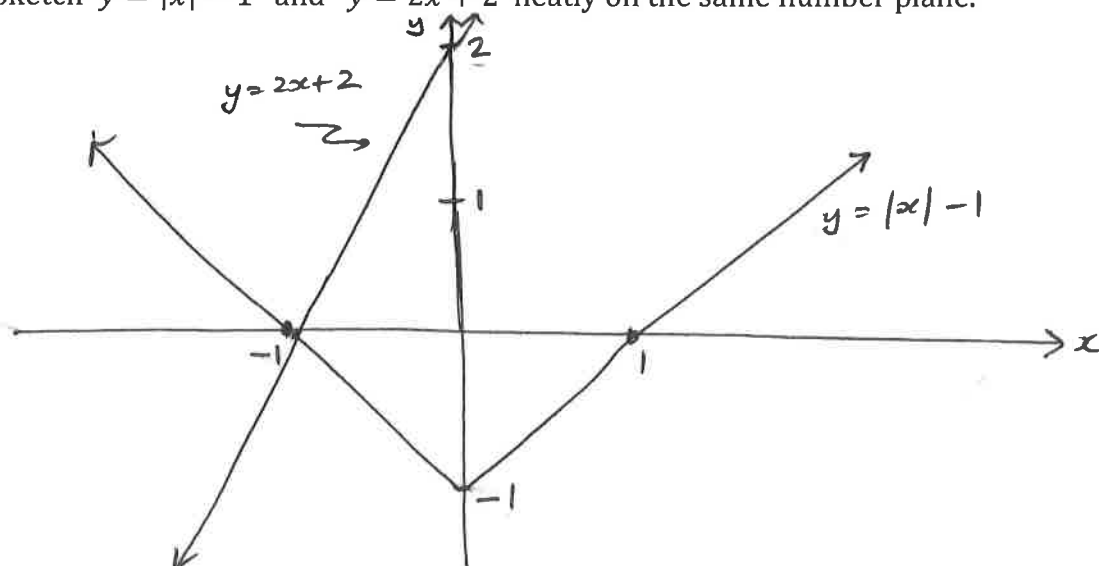
$$= -x + \frac{\pi}{2}$$

$$\therefore y = -x + \pi \quad (\text{i.e. } x + y - \pi = 0)$$

Question 36 (2 marks)

(a) Sketch $y = |x| - 1$ and $y = 2x + 2$ neatly on the same number plane.

1



(b) Hence solve the equation $|x| - 2x = 3$

1

$$\text{If } |x| - 2x = 3 \text{ then } |x| = 2x + 3$$

$$\therefore |x| - 1 = 2x + 2$$

So solution will be where graphs intersect in part (a)

$$\therefore x = -1$$

Question 37 (4 marks)

One half percent (0.5 %) of a country has a certain viral disease. A test is developed for the disease. The test gives a false positive 3% of the time, and a false negative 2% of the time.

- (a) Show that the probability that Andy, a randomly selected person, tests positive is 0.03475

2

[Hint: in this question, let D be the event that Andy has the disease, and $\bar{D}$ be the event Andy does not have it. Let T be the event that Andy's test comes back positive.]

$$\begin{aligned} P(T) &= P(T|D) \cdot P(D) + P(T|\bar{D}) \cdot P(\bar{D}) \\ &= (0.98)(0.005) + (0.03)(0.995) \\ &= 0.03475 \end{aligned}$$

✓✓

- (b) Andy just got the bad news that his test came back positive.

Find the probability that Andy actually has the disease.

2

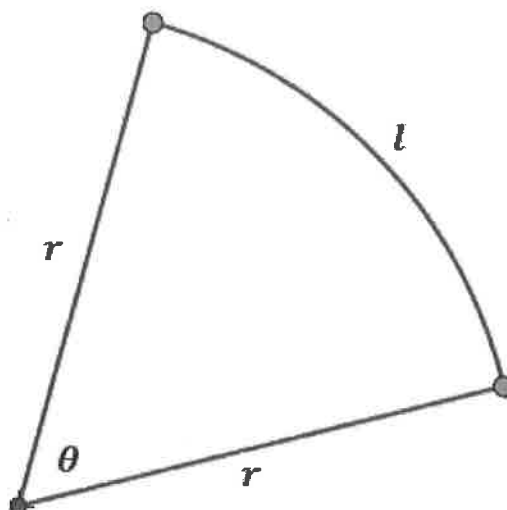
$$\begin{aligned} P(D|T) &= \frac{P(D \cap T)}{P(T)} = \frac{P(T \cap D)}{P(T)} \\ &= \frac{P(T|D) \cdot P(D)}{P(T)} \\ &= \frac{(0.98)(0.005)}{0.03475} \end{aligned}$$

$$\begin{aligned} &= 0.141 \dots \\ &\text{(i.e. } \hat{=} 14\%) \end{aligned}$$

✓✓

Question 38 (5 marks)

The diagram below shows a sector of a circle of radius r centimetres. The angle at the centre is θ radians, and the perimeter of the whole sector is 8 cm.



- (a) Show that $r = \frac{8}{2+\theta}$.

1

$$l = r\theta$$

$$\therefore r + r + r\theta = 8$$

$$2r + r\theta = 8$$

$$r(2+\theta) = 8$$

$$\therefore r = \frac{8}{2+\theta}$$

as required. ✓

- (b) Show that A , the area of the sector in cm^2 , is given by

$$A = \frac{32\theta}{(\theta+2)^2}$$

1

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}\left(\frac{8}{2+\theta}\right)^2 \cdot \theta = \frac{1}{2} \cdot \frac{64\theta}{(2+\theta)^2}$$

$$\therefore A = \frac{32\theta}{(2+\theta)^2}, \text{ as required. } \checkmark$$

- (c) If $0 \leq \theta \leq \frac{\pi}{2}$, find the maximum area of the sector, and the value of θ for which this occurs.

3

$$\frac{dA}{d\theta} = \frac{(\theta+2)^2 \cdot 32 - (32\theta) \cdot 2(\theta+2)}{(\theta+2)^4}$$

✓

For max. A, $\frac{dA}{d\theta} = 0 \quad \therefore 32(\theta+2)^2 - 64\theta(\theta+2) = 0$

$$\therefore 32(\theta+2)[\theta+2 - 2\theta] = 0$$

$$\therefore 32(\theta+2)(2-\theta) = 0$$

$$\therefore \theta = -2 \text{ or } \theta = 2$$

But $0 \leq \theta \leq \frac{\pi}{2}$, so these both lie outside the possible range of θ .

So we need to test the endpoints, 0 and $\frac{\pi}{2}$:-

At $\theta = 0$, $A = 0$

$$\text{At } \theta = \frac{\pi}{2}, A = \frac{32\left(\frac{\pi}{2}\right)}{\left(\frac{\pi}{2} + 2\right)^2} = \frac{16\pi}{\left(\frac{\pi+4}{2}\right)^2}$$

$$\therefore A = \frac{64\pi}{(\pi+4)^2}$$

$$\doteq 3.9422 \text{ cm}^2$$

Thus max. area = $\frac{64\pi}{(\pi+4)^2} \text{ cm}^2 \doteq 3.9422 \text{ cm}^2$,
obtained when $\theta = \frac{\pi}{2}$ radians.

✓✓

End of examination